

ALEX J. MAILE

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(Current as of : 22 April 2026)

EDUCATION

2023 — (Current) **Ph.D. Ecology and Evolution**

University of Kansas, Lawrence, Kansas
Graduate Advisor: Dr. W. Leo Smith

2023 — **M.S. Biological Sciences**

Thesis: **Evolutionary Relationships of the Anglerfishes (Lophioidei) with a Total Evidence Approach: Illuminating Habitat Transition Impacts on Anglerfish Lures and Body Shapes**
St. Cloud State University, St. Cloud, Minnesota
Graduate Advisor: Dr. Matthew P. Davis

2020 — **B.S. Biology: Biodiversity, Ecology, and Evolution** (Honors)

St. Cloud State University, St. Cloud, Minnesota
Undergraduate Advisor: Dr. Matthew P. Davis

PUBLICATIONS

- 2026 **Maile, A.J.** & Davis, M. P. 2026 The evolution of lures in anglerfishes (Acanthuriformes: Lophioidei): Investigating nature's tackle box. *Journal of Ichthyology & Herpetology* 114:103—118. <https://doi.org/10.1643/i2025018>
- 2025 **Maile, A.J.**, Smith, W.L., & Davis, M. P. 2025. A total-evidence phylogenetic approach to understanding the evolution, depth transitions, and body-shape changes in the anglerfishes and allies (Acanthuriformes: Lophioidei). *PLoS ONE* 20: e0322369. <https://doi.org/10.1371/journal.pone.0322369>
- 2021 Ghedotti, M. J., DeKay, H.M., **Maile, A.J.**, Smith, W.L., and Davis, M.P. 2021. Anatomy and evolution of bioluminescent organs in slimeheads (Trachichthyidae). *Journal of Morphology* 282: 820—832. <https://doi.org/10.1002/jmor.21349>
- 2020 **Maile, A.J.**, DeArmon, E.S, May, Z.A., Martin, R., & Davis, M.P. 2020. Marine habitat transitions and body shape evolution in lizardfishes and their allies (Aulopiformes). *Copeia* 108(4): 820—832. <https://doi.org/10.1643/CG-19-300> **[Best Student Paper in Copeia 2020]**

GRANTS AND SCHOLARSHIPS

2026	Dr. Bøje Benzon Foundation Grant (\$1570), Natural History Museum of Denmark
	Panorama Grant (\$3000), Biodiversity Institute and Natural History Museum
	Burke Museum Collections Study Grant (\$1,400), University of Washington
	Rinker Endowment PhD Graduate Student Research (\$500), University of Kansas
2025	NSF-funded BioGEM Research and Mentoring for Postbaccalaureates in Biological Sciences Program Summer Funding (\$5,000), University of Kansas
	Val and Marilyn Smith PhD Graduate Student Support Endowment Fund (\$350), University of Kansas
	Graduate Student Organization Travel Award (\$500), University of Kansas
2024	Martha Mitchell Pearson Scholarship (\$7300), University of Kansas
	Graduate Student Organization Travel Award (\$500), University of Kansas
	Val and Marilyn Smith PhD Graduate Student Support Endowment Fund (\$300), University of Kansas
2023	Student Research Travel Grant (\$500), St. Cloud State University
2022	George W. Friedrich Wildlife Protection Fund (\$3000), St. Cloud State University
	Dorothy Esler Memorial Scholarship (\$500), St. Cloud State University
	Natural History Museum of Los Angeles County Student Collections Study Award (\$1344), Natural History Museum of Los Angeles County
2021	Student Research Grant (\$962), St. Cloud State University
	Raney Grant (\$1000), American Society of Ichthyologists and Herpetologists
	George Friedrich Scholarship (\$1000), St. Cloud State University
2020	Student Research Grant (\$1000), St. Cloud State University
	Charles and Ellora Alliss Scholarship (\$1500), St. Cloud State University
	Harold and Gladys I. Hopkins Scholarship Fund Scholarship (\$1000), St. Cloud State University
	Jackie Hanson Environmental Studies Scholarship (\$200), St. Cloud State University
2019	SCSU Scholarship (\$1500), St. Cloud State University

- Student Research Grant** (\$1000), St. Cloud State University
- Huskies Summer Undergraduate Research Fellowship SURF** (\$500), St. Cloud State University
- 2018 **Harold and Gladys I. Hopkins Scholarship Fund Scholarship** (\$1000), St. Cloud State University
- Jackie Hanson Environmental Studies Scholarship** (\$200), St. Cloud State University
- Student Research Grant** (\$950), St. Cloud State University
- Huskies Summer Undergraduate Research Fellowship SURF** (\$500), St. Cloud State University
- 2017 **Avon Sportsmen's Club** (\$200), Albany High School

PROFESSIONAL PRESENTATIONS

- 2025 **Maile, A.J.** & Davis, M.P. A The Evolution of Lures in Anglerfishes (Acanthuriformes: Lophioidei): Investigating Nature's Tackle Box. Joint Meeting of Ichthyologists and Herpetologists, St. Paul, Minnesota, July 2025. Oral Presentation.
- 2024 **Maile, A.J.** Smith, W.L., & Davis, M.P. A Combined Morphological and Molecular Phylogenetic Analysis of the Anglerfishes (Acanthuriformes: Lophioidei). Joint Meeting of Ichthyologists and Herpetologists, Pittsburg, Pennsylvania, July 2024. Oral Presentation.
- 2022 **Maile, A.J.** & Davis, M.P. Evolutionary Relationships of Anglerfishes Based on Genomic and Morphological Data. Huskies Showcase, St. Cloud State University, St. Cloud, Minnesota, April 2022. Poster Presentation.
- 2020 **Maile, A.J.** & Davis, M.P. Evolution of Biofluorescence in Rabbitfishes (Siganidae). Huskies Showcase, St. Cloud State University, St. Cloud, Minnesota, April 2020. Poster Presentation.
- 2019 **Maile, A.J.** & Davis, M.P. Evolution of Biofluorescence in Rabbitfishes (Siganidae). Joint Meeting of Ichthyologists and Herpetologists, Snowbird, Utah, June 2019. Poster Presentation.
- Maile, A.J.** & Davis, M.P. Body Shape Changes Across Habitat Transitions in Lizardfishes (Aulopiformes). Huskies Showcase, St. Cloud State University, St. Cloud, Minnesota, April 2019. Poster Presentation.
- Maile, A.J.** & Davis, M.P. Body Shape Changes Across Habitat Transitions in Lizardfishes (Aulopiformes). Huskies Showcase, St. Cloud State University, St. Cloud, Minnesota, April 2019. Oral Presentation.
- 2018 **Maile, A.J.** & Davis, M.P. Body Shape Changes Across Habitat Transitions in Lizardfishes (Aulopiformes). Joint Meeting of Ichthyologists and Herpetologists, Rochester, New York, July 2018. Poster Presentation.

JOURNAL SERVICE

2024 — Current I have reviewed 7 manuscripts from the following 5 journals:
Ichthyology and Herpetology (3) *Biology Letters* (1)
California Fish and Wildlife Journal (1) *Frontiers in Zoology* (1)
ZooKeys (1)

PROFESSIONAL EXPERIENCE

- 2026 **Curatorial Assistant:** Ichthyology, University of Kansas, Lawrence, Kansas. Supervisor: Andy Bentley and Leo Smith.
- Imaged University of Kansas ichthyology type collection for Specify database.
- 2024, 2026 **Graduate Teaching Assistant:** Ichthyology (BIOL 592), University of Kansas, Lawrence, Kansas. Supervisor: Dr. W. Leo Smith
- Assisting undergraduate students with fish identification and gross anatomy dissections during laboratory sessions.
- 2025 **Graduate Teaching Assistant:** Tree of Life (BIOL 413), University of Kansas, Lawrence, Kansas. Supervisors: Drs. Brian Atkinson & Richard Glor
- 2025 **Graduate Teaching Assistant:** Introduction to Systematics (BIOL 428), University of Kansas, Lawrence, Kansas. Supervisor: Dr. W. Leo Smith
- 2024 **Curatorial Assistant:** Ichthyology, University of Kansas, Lawrence, Kansas. Supervisor: Andy Bentley.
- Identified and processed freshwater Pflieger fish collection.
- 2024 **Graduate Teaching Assistant:** Principles of Biology (BIOL 100), University of Kansas, Lawrence, Kansas. Supervisor: Dr. Trevor Rivers.
- Grading online assignments and discussions.
- 2023 **Graduate Teaching Assistant:** Introduction to Biostatistics (BIOL 370), University of Kansas, Lawrence, Kansas. Supervisor: Dr. Maggie Wagner and Dr. Tara Marriage.
- Assisting undergraduate students with lecture problems, supervising multiple lab sessions, and help students utilize R to compute and demonstrate statistical concepts for lab questions.
- 2023 **Graduate Media Production Assistant,** Innovating the Science Classroom for Widespread Student Engagement, St. Cloud, Minnesota. Supervisor: Dr. Matthew P. Davis
- Collaborating with experts in producing high-quality short scientific videos for open-source access used specifically toward targeting underrepresented STEM groups and provide digital and technological equity to students from various backgrounds.
- 2022 — 2023 **Graduate Teaching Assistant:** Genetics (BIOL 262), St. Cloud State University, St. Cloud, Minnesota. Supervisor: Dr. Christopher Kvaal
- Assisting and lead genetics laboratories including practicing DNA techniques, PCR, and exercises related to evolution and agriculture.

- 2021 — 2022 **Graduate Teaching Assistant:** Cell Biology and Function (BIOL 151), St. Cloud State University, St. Cloud, Minnesota. Supervisor: Dr. Michelle Wagner
- Supervised introductory biology laboratories including coordinating of student lab scheduling, grading, and running labs involving cell biology, inheritance, and function.
- 2016-2021 **Invertebrate Aqua culturist and Custom Aquaria 3D Designer**, Maile Made Reef Supplies, Avon, Minnesota. **Personal Business**
- Production of invertebrate aquaculture products to be sold directly to consumers and aquarium stores in the Twin Cities of Minnesota.
 - Consult, collaborate, and design aquaria supplies and equipment through 3D modeling and 3D printing by custom request from direct consumers and aquarium stores in the Twin Cities of Minnesota.
- 2019 **Huskies Summer Undergraduate Research Fellowship**, St. Cloud State University, St. Cloud, Minnesota. Supervisor: Dr. Matthew P. Davis.
- Studying the evolution of color and pattern changes in deep-sea fishes.
- 2017 — 2020 **Undergraduate Researcher**, St. Cloud State University, St. Cloud, Minnesota. Supervisor: Dr. Matthew P. Davis
- Studying the evolution of body shape changes and habitat transitions in Lizardfishes (Aulopiformes).
 - Studying biofluorescence and venom association in marine Rabbitfishes (Siganidae).
- 2018 — 2019 **Undergraduate Lab Assistant**, Xcel Energy Larval Fishes Identification Project, St. Cloud State University, St. Cloud, Minnesota. Supervisor: Dr. Matthew P. Davis
- Identifying larval and juvenile fishes of the Mississippi River in Minnesota contributing to Xcel Energy's Biological Impact Assessment of water intake from coal plants.
- 2016 — 2018 **AIS Watercraft Inspector Level 2**, Lamb Labor Services, Paynesville, Minnesota
- On site coordinator for removal and decontamination of watercraft at multiple public lake access points. Work also involving coordinating AIS Inspector Level 1s in large scale fishing events requiring watercraft decontamination.

PROFESSIONAL SERVICE AND ASSOCIATIONS

- 2026 — Current **Member of the American Society of Naturalists**
- 2025 — Current **Treasurer of the Ecology and Evolutionary Biology Graduate Student Organization**, University of Kansas.
- 2025 — Current **Chair: Design Committee of the Ecology and Evolutionary Biology Graduate Student Organization**, University of Kansas.
- 2024 — 2025 **Co-Mentor, NSF Research and Mentoring for Postbaccalaureates in Biological Sciences (RaMP): BioGEM Program**, University of Kansas
- Co-mentoring postbaccalaureate scholar in the NSF-funded "Integrating Biodiversity patterns with Genetic Evolutionary Mechanisms (BioGEM)" program focused on research training and professional development.

2023 — Current	Member of the Ecology and Evolutionary Biology Graduate Student Organization, University of Kansas.
2022 — 2023	Secretary of the Biology Graduate Student Association (BGSA), St. Cloud State University Mentoring BGSA newly elected officers while striving to regain and increase graduate student participation and membership.
2022	Best Ichthyological Papers in Ichthyology and Herpetology Committee Requested in helping to select the best papers in Ichthyology and Herpetology. Evaluated publications in Ichthyology and Herpetology (formerly Copeia) to rank the papers according to the categories of Best Paper Overall, Best Paper Young Scholar, and Best Student Paper.
2021 — 2022	President of the Biology Graduate Student Association (BGSA), St. Cloud State University - Leading in developing a comprehensive strategy focused on rejuvenating a biological graduate student organization stifled by COVID 19 to help ensure the organization's short and long-term objectives and success. - Assisted the Biochemistry and Molecular Biology club in designing a Research Open House to allow students to get first-hand experience in the levels and types of research conducted in the School of Science and Engineering.
2020 — 2021	President of Into the Wild Society (ITWS), St. Cloud State University Vice President of Into the Wild Society (ITWS), St. Cloud State University Founder of a student-led organization on campus dedicated to creating a diverse and inclusive community of individuals with a strong interest in ecology and evolution while organizing workshops and networking with experts of their field.
2019 — 2020	COSE Strategic Planning Committee, St. Cloud State University - Articulating the needs and concerns of biological students to collaborate with faculty in creating a transformative educational strategic plan that responds to students, our society, and the world. - Discussed with department heads of the School of Science and Engineering about the implementation of required student-teacher evaluations.
2018 — 2020	Dean's Student Advisory Board, St. Cloud State University The student nominee for the department of biological sciences tasked with creating a direct line of communication voicing the needs and concerns of unvoiced students directly to the dean of the school of science and engineering.
2017 — Current	Member of American Society of Ichthyologists and Herpetologist

AWARDS AND HONORS

2021 Copeia 2020: **Best Student Paper in Ichthyology**

- See Maile et al., 2020. First undergraduate ever selected as Best Student Paper in Ichthyology by the American Society of Ichthyologists and Herpetologists

2019-2020

Deans Student List, Spring Semester & Fall Semester

2019

Best Our Husky Compact Reflection, Husky Showcase